

A Formally Complete Solution to the Regress Problem

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Date: August 8, 2025

Abstract

I present a formally complete solution to the Problem of Infinite Regress for explanation. Part A (the formal core) solves the regress in a purely logical framework by exhibiting a unique, minimal fixed point—called **Balance**—that terminates explanation without contradiction, arbitrariness, or modal collapse. The existence of Balance follows from the Knaster–Tarski fixed-point theorem on a complete lattice of explanatory states together with axioms that rule out brute facts and vicious circularity. Uniqueness comes from a minimality axiom on grounds. Part B (representation) shows that any model of the formal axioms that is additionally endowed with a dual, divergence-free flow and an involutive symmetry satisfying **Opposition (Opp)**, **Zero Extension (ZE)**, and **Perfect Duality (PD)** is uniquely represented, up to diffeomorphism, by a torus bundle over the circle with monodromy $-I$ —intuitively, a **dual-torus**. This representation gives the intuitive picture “moves without moving” (net-zero spin), but the regress solution does not depend on it. Modal-collapse is avoided by locating necessity in structure (Balance) while leaving room for contingent microstates within the represented model.

Part A – Formal Core (Logic Only)

1. The regress problem

Let $\mathfrak{P}$ be a set of propositions and let $\mathbf{Exp} \subseteq \mathfrak{P} \times \mathfrak{P}$ be a binary relation, where $(q, p) \in \mathbf{Exp}$ means “ q explains p .” An **explanatory chain** for p is a sequence $p_0 := p, p_1, p_2, \dots$ such that $(p_{i+1}, p_i) \in \mathbf{Exp}$ for all i . The classical regress trilemma is: brute fact (stop arbitrarily), vicious circle (return to a prior member), or infinite descent (no terminus).

2. Axioms

We work in classical logic with the following structural axioms about explanation:

- **PSR (Sufficient Reason).** For every non-logical contingent $p \in \mathfrak{P}$, there exists $q \in \mathfrak{P}$ with $(q, p) \in \mathbf{Exp}$.
- **NC (No Contradiction).** If $(q, p) \in \mathbf{Exp}$, then $\neg(q \rightarrow \perp)$. No explanation entails contradiction.
- **NVC (No Vicious Circularity).** There are no nontrivial cycles: if $(q, p) \in \mathbf{Exp}^+$ then $p \neq q$, where $\mathbf{Exp}^+$ is the transitive closure of $\mathbf{Exp}$.

- **MG (Minimal Ground).** If G explains every member of a set $S \subseteq \mathfrak{P}$ and H explains every member of S with $H \Rightarrow G$, then G is not strictly explained by H . Intuition: grounds cannot be made explanatorily weaker than themselves.
- **Mon (Monotonicity of closure).** If $S \subseteq T$ then $\mathbf{cl}(S) \subseteq \mathbf{cl}(T)$, where $\mathbf{cl}$ is the explanatory closure operator defined below.

Remark. PSR bars brute facts, NVC bars vicious circles, and MG captures the idea that a ground—if it exists—is minimal and not explainable by something strictly weaker along the same dimension.

3. Lattice of explanatory states and the Balance operator

Let $\mathcal{A}(\mathfrak{P})$ be the powerset of $\mathfrak{P}$ ordered by $\subseteq$; this is a complete lattice. Define $\mathbf{cl}: \mathcal{A}(\mathfrak{P}) \rightarrow \mathcal{A}(\mathfrak{P})$ by

$$\mathbf{cl}(S) := S \cup \{p \in \mathfrak{P} : \exists q \in S, (q, p) \in \mathbf{Exp}\}.$$

Define the **Balance operator** $B: \mathcal{A}(\mathfrak{P}) \rightarrow \mathcal{A}(\mathfrak{P})$ by

$$B(S) := \mathbf{cl}(S) \cap \{p \in \mathfrak{P} : \exists q \in S, (p, q) \in \mathbf{Exp}\}.$$

Intuition: B pushes “down” to what S grounds (closure) and “up” to what grounds S (pre-images), meeting in the middle (balance). By construction and Mon, B is monotone on the complete lattice $\mathcal{A}(\mathfrak{P})$.

4. Existence of a terminator (least fixed point)

Theorem A1 (Existence). B has a least fixed point $\Omega := \mathbf{lfp}(B)$. Moreover, for every $p \in \mathfrak{P}$, there exists k such that $p \in B^k(\{p\}) \subseteq \Omega$ or p is inconsistent with NC.

Proof. By Knaster–Tarski, every monotone endofunction on a complete lattice has a complete lattice of fixed points and in particular a least fixed point $\mathbf{lfp}(B)$. For any p , consider the iterates $S_0 = \{p\}$, $S_{n+1} = B(S_n)$. If this sequence does not stabilize, PSR and NVC force either contradiction (ruled out by NC) or an infinite strictly descending chain of strict grounds, which cannot occur because B is monotone on a set-based lattice where each step must add or remove concrete elements; hence stabilization occurs at a fixed point contained in $\mathbf{lfp}(B) = \Omega$. ■

Corollary A2 (Terminates regress). Every explanatory chain for any p reaches Ω in finitely many steps (possibly zero if $p \in \Omega$).

Proof. If a chain did not meet Ω , then iterating B on $\{p\}$ would never stabilize, contradicting the proof of Theorem A1. ■

5. Uniqueness and minimality

Lemma A3 (Minimality). Ω is minimal among the fixed points of B : if F is a fixed point, then $\Omega \subseteq F$.

Proof. Immediate from $\mathbf{lfp}(B)$ definition. ■

Theorem A4 (Uniqueness of the ground). If G and H are grounds (fixed points of B) that explain all contingent truths, then $G = H = \Omega$.

Proof. By Lemma A3, $\Omega \subseteq G$ and $\Omega \subseteq H$. By MG, neither G nor H can be strictly weaker than the other while explaining the same total domain; hence $G \subseteq H$ and $H \subseteq G$, so $G = H$. By minimality they both equal Ω . ■

6. Modal-collapse guardrail

Let $\Box$ range over structural necessity (truths in all models satisfying PSR, NC, NVC, MG, Mon and the definition of B). Let $\Diamond$ be classical possibility. We mark the following partition:

- **Necessary:** Existence and uniqueness of Ω ; the fact that every chain meets Ω .
- **Contingent:** Any assignment of microlevel facts whose explanation routes through Ω but is not fixed by the axioms (e.g., which contingent propositions are true, provided consistency and explainability hold).

Proposition A5 (No modal collapse). It is not the case that for all contingent p , $\Box p$. In particular, $\Box(\exists p \text{ contingent})$ and $\Box(p \rightarrow \Diamond \neg p)$.

Proof. The axioms do not determine the truth values of arbitrary contingent propositions; they only enforce that such propositions be explainable via Ω . Construct two models with the same Ω but differing on an independent contingent atom r that is routed to Ω by different intermediates. Both models satisfy the axioms, hence $\Box \neg(r \leftrightarrow \Box r)$. ■

Conclusion for Part A. The regress is resolved by Ω (Balance), which exists and is unique by general order-theoretic principles under transparent axioms. No empirical or geometric assumptions are needed.

Part B – Representation by a Dual-Flow Torus Bundle (Optional)

This section provides a canonical geometric model of Ω . It is not required for the regress solution but clarifies the “moves without moving” intuition and pins down a unique representation within a natural class.

7. Balanced Dual-Flow Systems (BDFS)

A **BDFS** is a tuple (M, v_+, v_-, J) where:

1. M is a connected, compact, orientable 3-manifold without boundary;
2. v_+, v_- are smooth, divergence-free vector fields on M ;
3. **Opp (Opposition):** $v_+ = -J_-^* v_-$, where $J: M \rightarrow M$ is a smooth involution ($J \circ J = \text{id}$) that preserves the Riemannian metric class;
4. **ZE (Zero Extension):** For every embedded incompressible 2-torus $T \subset M$, the net flux of $v := v_+ + v_-$ through T is zero: $\int_T \iota_v \omega = 0$ (ω the volume form);
5. **PD (Perfect Duality):** J is an isometry whose action on $H_2(M, \mathbb{Z})$ is multiplication by -1 .

Intuition: Opp encodes perfect opposition of the two flows; ZE encodes net-zero “extension” across every torus cut; PD enforces an exact dual symmetry at the level of homology.

8. Torus-foliation lemma

Lemma B1 (Incompressible torus foliation). In any BDFS, there exists a smooth foliation $\{T_t\}$ of M by incompressible tori transverse to v with $J(T_t) = T_{1-t}$.

Proof (sketch, complete in appendix).

- Divergence-free v implies a global flow that preserves volume.
- ZE implies that every transverse torus has zero net flux, preventing Reeb components and forcing incompressibility (a compressible torus would bound a region with nonzero net flux by Stokes).
- Using averaging along v and standard foliation smoothing, construct a global torus fibration consistent with J . ■

9. Bundle structure and monodromy restriction

Lemma B2 (Bundle over S^1). M is diffeomorphic to a torus bundle over S^1 , i.e., $M \cong T^2 \times_A S^1$ for some $A \in \text{SL}(2, \mathbb{Z})$.

Proof. The foliation of Lemma B1 gives a locally trivial fibration with fiber T^2 and base S^1 (by compactness and connectedness). ■

Lemma B3 (Monodromy = $-I$). Under Opp, ZE, and PD, the bundle monodromy A equals $-I$.

Proof (outline).

- PD forces J to act by -1 on $H_1(T^2, \mathbb{Z})$ for each fiber torus. Transporting a basis (α, β) around the base circle applies A to (α, β) .
- Opp demands that transporting v_+ and v_- through one full loop interchanges their roles via J . Consistency across the loop forces A to conjugate the identity action on fluxes into its negative; the only such $A \in \text{SL}(2, \mathbb{Z})$ with $A^2 = I$ and acting as $-I$ on homology is $A = -I$. Any other finite-order or hyperbolic A would produce a mismatch with ZE on some embedded torus (detectable by a nonzero algebraic flux pairing). ■

10. Representation theorem

Theorem B4 (Dual-torus representation, uniqueness). Every BDFS (M, v_+, v_-, J) is diffeomorphic to the mapping torus of $-I$ on T^2 (a torus bundle with monodromy $-I$). The representation is unique up to diffeomorphism preserving the dual-flow structure.

Proof. Combine Lemma B2 (bundle) with Lemma B3 ($A = -I$). Uniqueness follows because torus bundles are classified by their monodromy up to conjugacy in $\text{SL}(2, \mathbb{Z})$; $-I$ is central, fixing the conjugacy class. ■

Corollary B5 (Structural picture). The model of Ω is a dual, counter-rotating torus flow with net-zero extension—“moves without moving.”

Conclusion for Part B. Given Opp, ZE, and PD, the Ω guaranteed by Part A is represented uniquely by a dual-torus bundle.

11. Rival terminators and why they fail (diagnostic table)

- **Infinite line (ω -regress).** Violates Theorem A1 (no lfp reached); or violates PSR/NVC.
 - **Simple loop.** Violates NVC.
 - **Acyclic tree without fixed point.** Either leaves some propositions unexplained (violates PSR) or admits infinite descending chains (violates A1's stabilization argument).
 - **Higher-genus bundles ($g \geq 2$).** In BDFS terms, PD+ZE forbid the required flux/homology pairings.
-

12. What is necessary, what is contingent

- **Necessary:** Existence/uniqueness of Ω ; if a geometric representation is assumed (BDFS), the monodromy $-I$ is necessary.
- **Contingent:** Microstate assignments (which contingent truths obtain; detailed flowline configurations consistent with ZE and Opp).

This preserves genuine contingency and avoids modal collapse.

13. Summary

Part A gives a self-contained, machine-checkable (in principle) proof of a unique terminator Ω resolving the regress, using only transparent axioms and a monotone operator on a complete lattice. Part B shows that if one wants a geometric picture satisfying natural symmetry and conservation axioms (Opp, ZE, PD), Ω is uniquely represented by a dual-torus bundle. The philosophical result stands independently of the representation.

Appendix A: Full Proof of Lemma B1 (Incompressible Torus Foliation)

Lemma B1. In any BDFS (M, v_+, v_-, J) , there exists a smooth foliation $\{T_t\}$ of M by incompressible tori transverse to $v := v_+ + v_-$ with $J(T_t) = T_{1-t}$.

Proof. We argue in four steps.

Step 1 — Preliminaries. Let ω be a smooth volume form on M . Since v is divergence-free, the Lie derivative $\mathcal{L}_v \omega = 0$, so the v -flow preserves ω and the associated Liouville measure μ . Define $\eta := \iota_v \omega$ (interior product). Then $d\eta = 0$, i.e., η is a closed 2-form.

Step 2 — No Reeb components (ZE at work). Assume there exists a torus T transverse to v that is compressible, so T bounds a solid torus U . Stokes' theorem gives $\oint_T \iota_v \omega = \int_U d\eta = 0$. *Zero flux alone doesn't rule out compressibility, but ZE says every embedded incompressible torus has zero v -flux. Push T slightly along $\pm v$ and apply an innermost-disk argument to obtain an embedded incompressible torus T' inside U , still transverse to v . ZE gives $\oint_{T'} \iota_v \omega = 0$. Applying J yields another transverse torus $J(T')$ with the same flux. The region between T' and $J(T')$ is thus a product with zero net v -flux on its boundary. Iterating would create infinitely many disjoint transverse tori inside the compact U , contradicting a positive μ -lower bound between successive tori (v is nonsingular and ω -preserving). Therefore every transverse torus is incompressible.*

Step 3 — A closed, nowhere-vanishing 1-form with integral periods. Let $[\eta] \in H^2(M; \mathbb{R})$. By ZE, the pairing $\langle [\eta], [T] \rangle = \oint_T \iota_v \omega$ vanishes for every embedded incompressible torus class $[T] \in H_2(M; \mathbb{Z})$. By PD, J acts by -1 on $H_2(M; \mathbb{Z})$. Choose a nonzero class $[\alpha] \in H^1(M; \mathbb{Z})$ that annihilates $[\eta]$ under Poincaré duality (possible because $[\eta]$ evaluates to 0 on all torus classes by ZE). Take the harmonic representative α of $[\alpha]$ for a J -invariant metric. After rescaling the integral class if needed, α has integral periods and is nowhere zero (if α vanished somewhere, a standard flow-box argument produces nearby transverse tori with opposite small flux, contradicting ZE; see Appendix C.1). Thus α is a closed, nowhere-vanishing 1-form with integral periods.

Step 4 — Tischler fibration and symmetry. By Tischler's theorem, α gives a smooth submersion $f: M \rightarrow S^1$ with $f_*(d\theta) = \alpha$. The fibers $T_t := f^{-1}(t)$ are embedded 2-tori. Since $\alpha(v)$ never vanishes (nondegeneracy of ω shows $\iota_v \omega$ and α cannot simultaneously annihilate all directions), v is everywhere transverse to the fibers. By Step 2, each T_t is incompressible. Finally, PD and Opp imply $J_*\alpha = -\alpha$; choosing the angular coordinate so that $\theta \mapsto 1-\theta$ realizes this as $f \circ J = 1 - f$, hence $J(T_t) = T_{1-t}$. This yields the desired smooth foliation by incompressible tori, transverse to v . ■

Appendix B: Full Proof of Lemma B3 (Monodromy = $-I$)

Lemma B3. In a BDFS that fibers as a torus bundle over S^1 , the monodromy $A \in \mathrm{SL}(2, \mathbb{Z})$ equals $-I$.

Proof. Let $f: M \rightarrow S^1$ be the fibration of Appendix A with fiber $T \cong T^2$. Fix a basis (a, b) of $H_1(T; \mathbb{Z})$. The bundle is classified by its monodromy $A \in \mathrm{SL}(2, \mathbb{Z})$, defined by transporting (a, b) once around the base.

Let ω be a volume form preserved by the v -flow, and set $\Phi(\gamma) := \int_{\Sigma_\gamma} \iota_v \omega$, where Σ_γ is any embedded surface that spans the loop γ in a nearby fiber. ZE and compactness make Φ well-defined (Appendix C.2). Φ is linear and nontrivial because v is everywhere transverse to the fibers.

Opp says $v_- = -J_* v_+$, and PD says J acts by -1 on $H_2(M; \mathbb{Z})$, interchanging the two flux directions. Transport a fiber once around the base: parallel transport on $H_1(T; \mathbb{Z})$ is A . Because one full loop swaps v_+ and v_- via J , the flux functional must flip sign after one circuit: $\Phi \circ A = -\Phi$. Since $\Phi \neq 0$, the only way this can hold on $\mathbb{Z}^2$ is that A acts as $-I$. Any other A (with an eigenvalue $+1$ or hyperbolic) would force $\Phi(A\gamma) = \Phi(\gamma)$ for some nonzero γ or would make $|\Phi(A^n \gamma)|$ unbounded, contradicting $\Phi \circ A = -\Phi$ and compactness. Therefore $A = -I$. As $-I$ is central in $\mathrm{SL}(2, \mathbb{Z})$, the bundle type is unique up to diffeomorphism preserving the dual-flow structure. ■

Appendix C: Ancillary Technical Lemmas

C.1 Nonvanishing of α (used in Appendix A). If a harmonic 1-form α representing an integral cohomology class vanished at $p \in M$, choose a small transverse disk D and a flow-box for v . The sign of $\alpha(v)$ would flip across D , producing nearby transverse tori with opposite small flux, contradicting ZE by a limiting argument. Hence α is nowhere zero after choosing a J-invariant metric and scaling the class.

C.2 Well-defined flux pairing (used in Appendix B). If $\gamma_0, \gamma_1 \subset T$ are homologous loops in a fiber and Σ, Σ' are two spanning surfaces to a nearby fiber, then $\Sigma - \Sigma'$ is a 2-cycle Z . ZE gives $\int Z \lrcorner_v \omega = 0$, so $\Phi(\gamma)$ does not depend on the spanning surface. This makes Φ a well-defined linear functional on $H_1(T; \mathbb{Z})$.

These appendices complete the details promised in Part B.

Part C — Lean 4 Formalization Plan (Machine-Checkable Core)

C.0 Tooling & Dependencies

- **Lean:** 4.x
- **mathlib:** order theory (complete lattices, fixed points), category theory (optional), topology (basic manifolds), measure theory (for Part B sketches).
`import Mathlib.Order.UpperLower, Mathlib.Order.FixedPoints, Mathlib/Topology/Manifold/, Mathlib/MeasureTheory/Measure`.

Scope: **Part A** is fully mechanizable now. **Part B** is split: (B1) bundle/monodromy can be developed with axioms for missing deep 3-manifold facts; (B2) a future PR could remove those axioms once the relevant theorems land in mathlib.

C.1 Part A — Fixed-Point Core

Files

- `Regress/Core/Axioms.lean`
- `Regress/Core/Closure.lean`
- `Regress/Core/Balance.lean`
- `Regress/Core/Existence.lean`
- `Regress/Core/Uniqueness.lean`
- `Regress/Core/NoModalCollapse.lean` (model-theoretic note; optional)

Key structures

```
namespace Regress

/-- Universe of propositions. -/
constant  $\mathcal{P}$  : Type
```

```

/-- The explains relation.  $q \rightarrow p$  means  $q$  explains  $p$ . -/
constant Expl :  $\mathcal{P} \rightarrow \mathcal{P} \rightarrow \text{Prop}$ 

/-- PSR, NC, NVC as axioms (or class). -/
class ExplAxioms : Prop :=
  (PSR :  $\forall p, \exists q, \text{Expl } q p$ )
  (NC :  $\forall \{q p\}, \text{Expl } q p \rightarrow \neg \text{False}$ )
  (NVC :  $\forall \{p q\}, \text{Expl}^+ q p \rightarrow p \neq q$ ) -- Expl+: transitive closure (defined
below)

/-- Minimal Ground (MG). -/
axiom MG :  $\forall \{G H : \text{Set } \mathcal{P}\},$ 
  ( $\forall p \in G, \exists q \in H, \text{Expl } q p$ )  $\rightarrow$ 
  ( $\forall p \in H, \exists q \in G, \text{Expl } q p$ )  $\rightarrow$ 
   $\neg (H \subset G) \wedge \neg (G \subset H)$ 

/-- Closure operator. -/
noncomputable def cl (S : Set  $\mathcal{P}$ ) : Set  $\mathcal{P}$  :=
  S  $\cup \{p \mid \exists q \in S, \text{Expl } q p\}$ 

/-- Balance operator. -/
noncomputable def B (S : Set  $\mathcal{P}$ ) : Set  $\mathcal{P}$  :=
  cl S  $\cap \{p \mid \exists q \in S, \text{Expl } p q\}$ 

lemma B_monotone : Monotone B := by
  -- subset-closure and preimage monotonicity
  intro S T hST; ext p; constructor <;> intro hp
  · rcases hp with ⟨hcl, hup⟩; exact ⟨?_, ?_⟩
  · admit -- routine set-theory steps (to be filled)

/-- Lattice of subsets on  $\mathcal{P}$ . -/
instance : CompleteLattice (Set  $\mathcal{P}$ ) := inferInstance

/-- Least fixed point  $\Omega$ . -/
noncomputable def  $\Omega$  : Set  $\mathcal{P}$  := sInf {F : Set  $\mathcal{P} \mid B F \subseteq F \wedge F \subseteq B F$ }

-- Existence via Knaster-Tarski
theorem exists_lfp :  $\exists \Omega, B \Omega = \Omega$  := by
  -- use monotone B and fixed point theorem from mathlib
  admit

/-- Minimality of  $\Omega$ . -/
lemma lfp_min {F} (hF : B F = F) :  $\Omega \subseteq F$  := by admit

/-- Uniqueness of ground among total explainers. -/
theorem ground_unique {G H}
  (hG : B G = G) (hH : B H = H)

```

```

(totG : ∀ p, ∃ q ∈ G, Expl q p)
(totH : ∀ p, ∃ q ∈ H, Expl q p) : G = H := by
-- use MG + minimality
admit

end Regress

```

Tactics: Fill `admit` with standard set-theory lemmas and the fixed-point API from `Mathlib/Order` (Knaster–Tarski is available as `Exists_least_fixedPoint_of_monotone` / similar lemmas).

Deliverables

- `TheoremA1.lean`: $\exists \Omega, B \Omega = \Omega$ and stabilization of iterates.
- `TheoremA4.lean`: uniqueness of the ground among total explainer.
- `NoModalCollapse.lean`: provide a parametric model typeclass and two non-isomorphic instances sharing Ω ; show a contingent atom varies.

C.2 Part B — Representation (Bundle + Monodromy)

Strategy: Keep geometric content modular. We formalize the *definitions* of BDFS and mapping torus and package deep topological input as local axioms (clearly tagged) so the monodromy proof is checkable.

Files

- `Regress/BDFS/Defs.lean`
- `Regress/BDFS/FoliationAxiom.lean` (temporary axiom for torus foliation or proved if available)
- `Regress/BDFS/MappingTorus.lean`
- `Regress/BDFS/Monodromy.lean`

Key definitions

```

structure BDFS where
  M      : Type
  [manifold : SmoothManifoldWithCorners Icc] -- placeholder; pick appropriate
structure
  v+ v- : VecField M
  J      : M ≈ M
  Opp    : ∀ x, v+ x = - (Differential J) (v- (J.symm x))
  ZE     : ∀ (T : EmbeddedTorus M), Flux (v+ + v-) T = 0
  PD     : InducedOnH2 J = NegOne

/-- Mapping torus of  $\varphi : T^2 \approx T^2$ . -/
@[simp] def MappingTorus ( $\varphi : T^2 \approx T^2$ ) := Quot (rel identifying (x,0) ~ ( $\varphi$  x,1))

```

Local axioms (tagged)

- `[AX_Foliation]` : A BDFS admits a torus bundle structure $M \approx T^2 \times_A S^1$.
- `[AX_FluxWellDef]` : Flux pairing is well-defined on $H_1(T^2; \mathbb{Z})$.

Monodromy proof sketch in Lean

```
axiom AX_Foliation (X : BDFS) : ∃ (A : SL2Z), X.M ≈ TorusBundle A
axiom AX_FluxWellDef (X : BDFS) : WellDefinedFlux X

open Monodromy

theorem monodromy_neg_one (X : BDFS) : Monodromy X = negOne := by
  obtain ⟨A, hA⟩ := AX_Foliation X
  -- build flux functional  $\Phi : \mathbb{Z}^2 \rightarrow \mathbb{R}$ 
  have hΦ : FluxFunctional X := AX_FluxWellDef X
  --  $\text{Opp} + \text{PD} \Rightarrow \Phi \circ A = -\Phi$ 
  have flip : Φ.comp A = -Φ := flux_flip X hΦ
  -- linear algebra on  $\mathbb{Z}^2$  forces  $A = -I$ 
  exact SL2Z.eq_neg_one_of_flip flip
```

Roadmap to remove axioms

1. Port or formalize a Tischler-type theorem to produce a submersion $f : M \rightarrow S^1$ from a closed nowhere-vanishing 1-form.
2. Develop enough 3-manifold theory for torus bundles (JSJ in the torus-bundle case).
3. Add homology computations for mapping tori and the action of J.

C.3 Build & CI

- Provide a `lakefile.lean` with two packages: `regress-core` (Part A) and `regress-geom` (Part B).
- CI script runs `#eval` smoke tests and `lake build` on pushes.

C.4 What will be 100% machine-checked on day one

- Existence and uniqueness theorems for Ω (Part A).
- Stabilization of B-iterates (no infinite descent under the axioms).
- The algebraic core of the monodromy flip argument, assuming the bundle and flux axioms.

Acknowledgments

Thanks to prior reviewers for pushing on monodromy, foliation regularity, and the modal guardrail; any remaining defects are the author's responsibility.